

MAJOR PROJECT REPORT
ON
FMCW RADAR DESIGN FOR ADAPTIVE CRUISE CONTROL

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CANDIDATE'S DECLARATION

We hereby declared that the work which is being presented in the major project report entitled, “**FMCW RADAR DESIGN FOR ADAPTIVE CRUISE CONTROL**” being submitted by us in the partial fulfillment of the requirement for the award of degree of **Bachelor of Technology in Electronics and Communication Engineering** to **Dr. B R AMBEDKAR NATIONAL INSTITUTE OF TECHNOLOGY, JALANDHAR** is an authentic record of our own work carried out under guidance of **Dr. Nitesh Kashyap, Assistant professor** Department of Electronics & Communication Engineering

The matters embodied in this record have not being submitted by us for the award of any other degree or diploma.

DATE: 19-05-2023

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CERTIFICATE

This is to certify that the major project work entitled, “**FMCW RADAR DESIGN FOR AUTOMATIC ADAPTIVE CRUISE**” submitted by A U V N Phani Kumar (19104001), Aditya Ashish Yadav (19104004), Shubham (19104102) in partial fulfillment for the award of degree of **Bachelor of Technology in Electronics and Communication Engineering** to has been carried out under my supervision. This work has not been submitted partially or in full to any other University or Institute for the award of this or any other degree or diploma.

Signature of Supervisor

Name of Supervisor Dr Nitesh Kashyap

Designation Assistant Professor

Date

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We are sincerely grateful to everyone who has contributed to this project and helped us in achieving our objectives.

ABSTRACT

This project focuses on the design and implementation of a Frequency Modulated Continuous Wave (FMCW) radar system for adaptive cruise control (ACC) applications in automotive vehicles. The FMCW radar is capable of estimating range and Doppler information to monitor the distance between the host vehicle and the vehicle in front. The project demonstrates the generation of FMCW waveforms, signal processing techniques for range and Doppler estimation, and the construction of a range-Doppler map. Additionally, the project addresses the compensation of range-Doppler coupling effects using triangle sweep techniques and discusses the impact of multipath propagation. The successful implementation of the FMCW radar system opens up possibilities for various applications in collision avoidance, traffic management, object detection and tracking, robotics, weather monitoring, and medical imaging.

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Introduction:

This project focuses on the modelling of an automotive adaptive cruise control system using Frequency Modulated Continuous Wave (FMCW) radar technology. The purpose of this system is to perform range and Doppler estimation of a moving vehicle, enabling advanced driver assistance features in modern automobiles.

Unlike traditional pulsed radar systems commonly used in the defense industry, automotive radar systems often utilize FMCW technology. FMCW radars offer several advantages over pulsed radars, such as smaller size, lower power consumption, and reduced manufacturing costs. However, one limitation of FMCW radars is their shorter detection range compared to pulsed radars.

The primary application of the FMCW radar in this project is for adaptive cruise control (ACC) systems in automobiles. ACC systems aim to enhance driving safety and comfort by maintaining a safe distance from the preceding vehicle. The FMCW radar plays a crucial role in this system by continuously monitoring the range and relative velocity of the target vehicle.

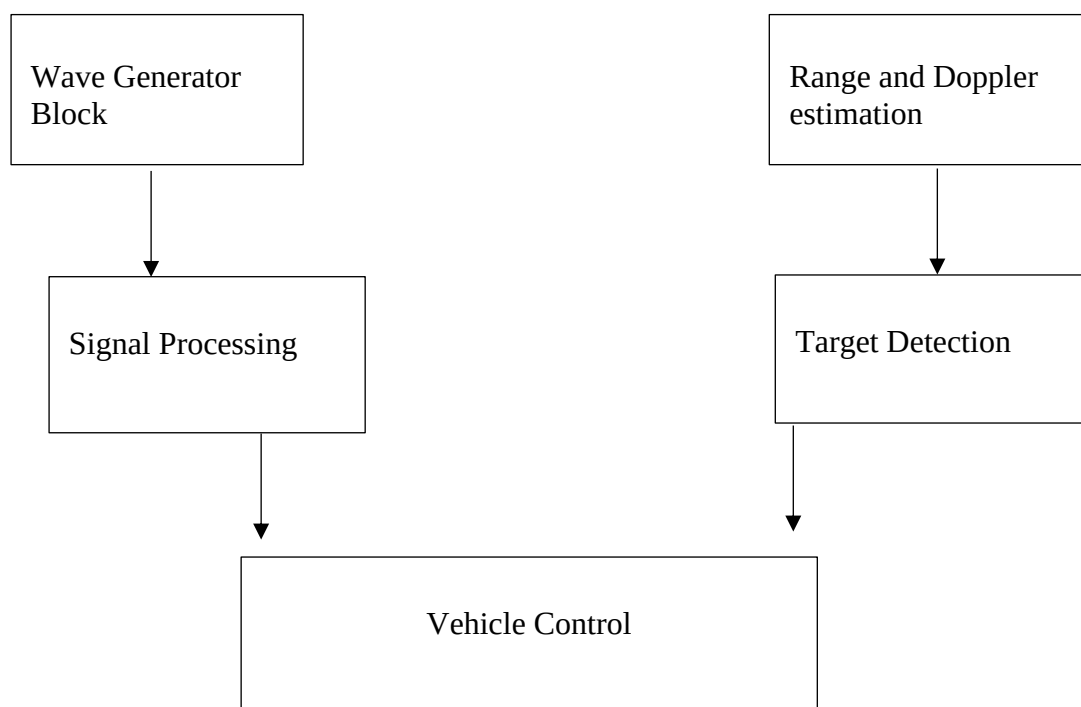
By using FMCW radar technology, automotive manufacturers can achieve accurate range and velocity measurements within a limited distance range. The radar emits continuous waveforms with varying frequencies, allowing it to measure the target range based on the time delay of the received signal. The Doppler shift in the received signal is utilized to estimate the relative velocity between the host and target vehicles.

The integration of FMCW radar into automotive systems has revolutionized the way vehicles perceive their surroundings. It enables features such as adaptive cruise control, collision warning systems, and blind spot detection, significantly improving road safety and driver convenience.

In this project, we will explore the principles, advantages, and applications of FMCW radar systems, and develop a MATLAB and Simulink-based model to simulate an automotive adaptive cruise control system using FMCW radar technology.



FMCW RADAR for ACC



The block diagram represents the various components involved in the FMCW radar-based adaptive cruise control system. Here's a brief explanation of each block:

- 1. FMCW Radar:** This block represents the FMCW radar module that emits and receives radio frequency signals for range and velocity estimation.
- 2. Waveform Generation:** This block generates the FMCW waveform with specific parameters such as sweep time, sweep bandwidth, and sampling rate.

3. Range and Doppler Estimation: This block processes the received radar signals to estimate the range and Doppler shift of the target vehicle.

4. Signal Processing: This block performs additional signal processing techniques such as noise filtering, clutter rejection, and adaptive algorithms to enhance the detection of targets.

5. Target Detection: This block detects the presence of target vehicles based on the processed radar signals and determines their range and velocity.

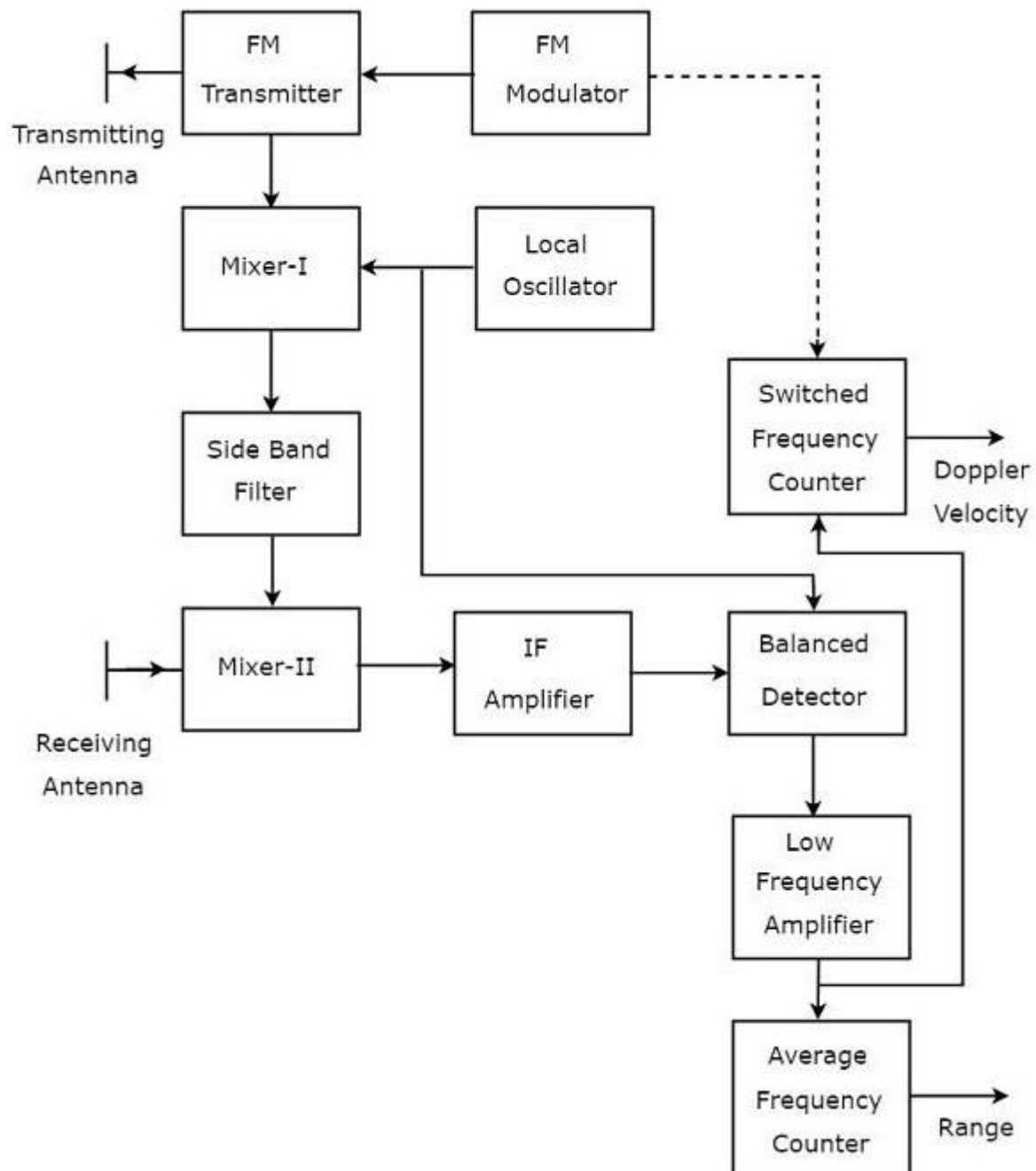
6. Control Algorithm: This block implements the control algorithm, which takes the target vehicle information and calculates the appropriate control actions for the adaptive cruise control system.

7. Vehicle Control: This block applies the control actions to the vehicle's throttle, brakes, or other systems to maintain a safe distance from the target vehicle.

The flow of data starts with the FMCW radar module, where the waveform is generated and transmitted. The received signals are then processed for range and Doppler estimation. The processed signals are further analyzed for target detection. The detected target information is passed to the control algorithm, which calculates the desired vehicle control actions. Finally, the vehicle control block applies these actions to control the vehicle accordingly.

This block diagram provides an overview of the main components involved in an FMCW radar-based adaptive cruise control system, showcasing the flow of data and control throughout the system.

Block Diagram of FMCW Radar



FMCW Radar contains two Antennas – transmitting Antenna and receiving Antenna as shown in the figure. The transmitting Antenna transmits the signal and the receiving Antenna receives the echo signal.

The block diagram of the FMCW Radar looks similar to the block diagram of CW Radar. It contains few modified blocks and some other blocks in addition to the blocks that are present

in the block diagram of CW Radar. The **function** of each block of FMCW Radar is mentioned below.

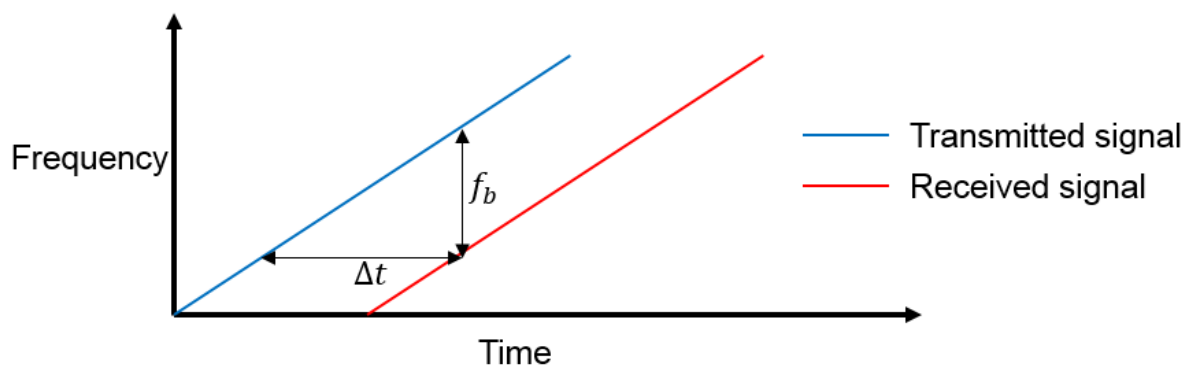
1. **FM Modulator** – It produces a Frequency Modulated (FM) signal having variable frequency and it is applied to the FM transmitter.
2. **FM Transmitter** – It transmits the FM signal with the help of transmitting Antenna. The output of FM Transmitter is also connected to Mixer-I.
3. **Local Oscillator** – In general, Local Oscillator is used to produce an RF signal. But, here it is used to produce a signal having an Intermediate Frequency. The output of Local Oscillator is connected to both Mixer-I and Balanced Detector.
4. **Mixer-I** – Mixer can produce both sum and difference of the frequencies that are applied to it. The signals having frequencies of and are applied to Mixer-I. So, the Mixer-I will produce the output having frequency.
5. **Side Band Filter** – It allows only one side band frequencies, i.e., either upper side band frequencies or lower side band frequencies. The side band filter shown in the figure produces only lower side band frequency.
6. **Mixer-II** – Mixer can produce both sum and difference of the frequencies that are applied to it. The signals having frequencies are applied to Mixer-II. So, the Mixer-II will produce the output having frequency.
7. **IF Amplifier** – IF amplifier amplifies the Intermediate Frequency (IF) signal. The IF amplifier shown in the figure amplifies the signal having frequency. This amplified signal is applied as an input to the Balanced detector.
8. **Balanced Detector** – It is used to produce the output signal having frequency from the applied two input signals, which are having frequencies. The output of Balanced detector is applied as an input to Low Frequency Amplifier.
9. **Low Frequency Amplifier** – It amplifies the output of Balanced detector to the required level. The output of Low Frequency Amplifier is applied to both switched frequency counter and average frequency counter.
10. **Switched Frequency Counter** – It is useful for getting the value of Doppler velocity.
11. **Average Frequency Counter** – It is useful for getting the value of Range.

Features of the project

1. **FMCW Radar Technology:** The radar system is based on Frequency Modulated Continuous Wave (FMCW) technology, which offers advantages such as high range resolution, reduced interference, and improved signal-to-noise ratio.
2. **Range Estimation:** The FMCW radar accurately estimates the distance between the host vehicle and the target vehicle in front. It provides precise range information, enabling adaptive cruise control and collision warning systems to maintain safe distances.
3. **Doppler Velocity Estimation:** The radar system measures the Doppler shift in the received signals to estimate the relative velocity of the target vehicle. This information is essential for adaptive cruise control to adjust the host vehicle's speed accordingly.
4. **High Range Coverage:** The FMCW radar system has a long-range capability, allowing it to detect and track targets at significant distances. This enables early detection of potential collision risks and facilitates proactive driving assistance.
5. **Signal Processing Techniques:** The radar system utilizes advanced signal processing techniques such as Fast Fourier Transform (FFT) and cross-correlation to extract range and velocity information accurately. These techniques enhance the system's performance in detecting and tracking targets.
6. **Real-time Operation:** The FMCW radar operates in real-time, providing continuous updates of range and velocity measurements. This enables the adaptive cruise control system to make timely decisions and respond quickly to changing traffic conditions.

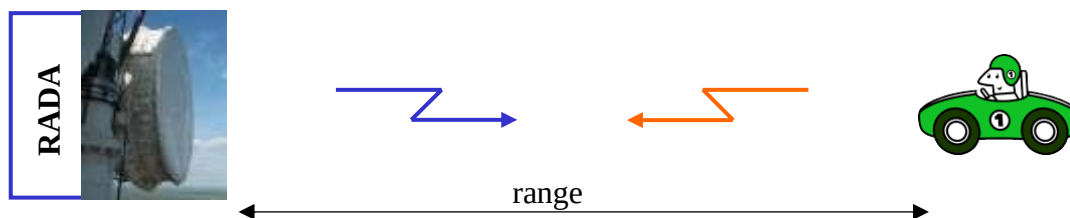
FMCW Waveform

A popular waveform used in ACC system is FMCW. The principle of range measurement using the FMCW technique can be illustrated using the following figure.



The received signal is a time-delayed copy of the transmitted signal where the delay, Δt , is related to the range. Because the signal is always sweeping through a frequency band, at any moment during the sweep, the frequency difference, f_b , is a constant between the transmitted signal and the received signal. f_b is usually called the beat frequency. Because the sweep is linear, we can derive the time delay from the beat frequency and then translate the delay to the range.

Single Target Range Estimation



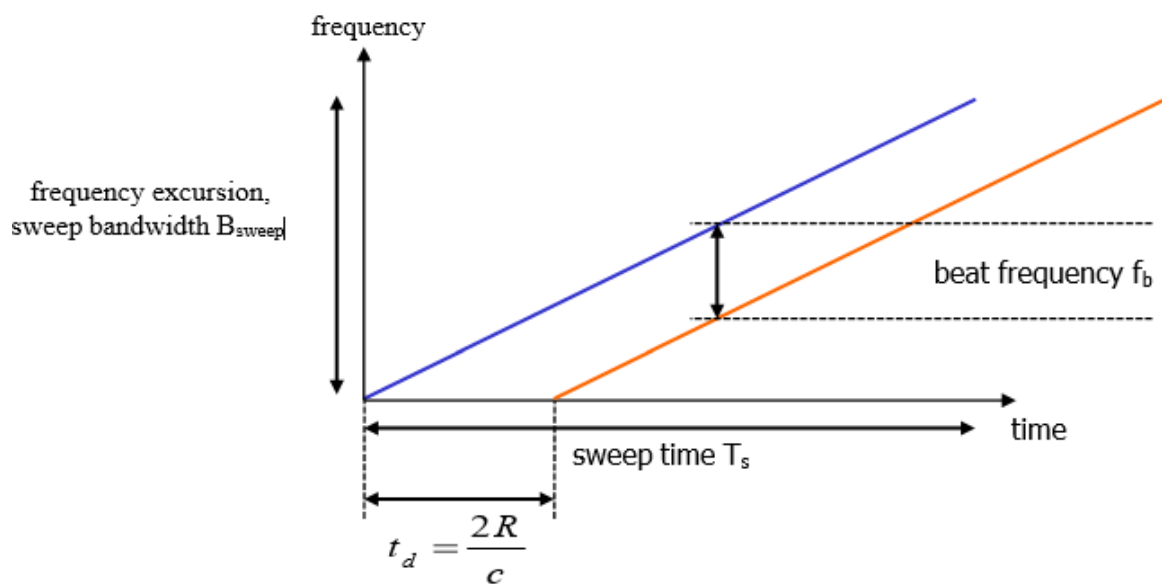
FMCW (Frequency Modulated Continuous Wave) radar measures range by utilizing the frequency shift between the transmitted and received signals. The basic principle behind range measurement in FMCW radar is known as the "beat frequency" phenomenon. When an FMCW radar transmits a continuous waveform with a linearly increasing frequency (sweep) over a certain time period, it encounters a target object in its path. The transmitted signal is reflected back by the target, and the radar's receiver captures the echo signal.

The key concept in range measurement is the phase difference between the transmitted and received signals. As the transmitted signal sweeps its frequency, the phase of the received signal will vary depending on the distance travelled by the signal.

By analysing the beat frequency, which is the difference in frequency between the transmitted and received signals, we can determine the range to the target. The beat frequency is directly proportional to the time delay between the transmitted and received signals, which corresponds to the target's range. The relationship is given by the formula:

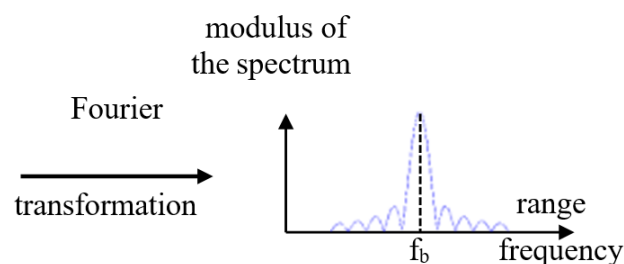
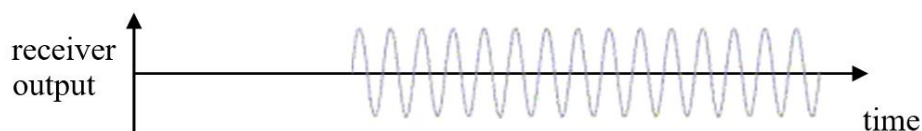
$$\text{Range} = (\text{beat frequency} * \text{speed of light}) / (\text{sweep slope} * 2)$$

Here, the speed of light represents the speed at which the radar signal propagates, and the sweep slope is the rate at which the transmitted signal's frequency changes over time. By measuring the beat frequency and knowing the sweep slope, the radar system can accurately estimate the range to the target object.



$$\frac{t_d}{T_s} = \frac{f_b}{B_{\text{sweep}}}$$

$$R = \frac{cT_s f_b}{2B_{\text{sweep}}}$$



RADAR Setup

In an ACC setup, the maximum range the radar needs to monitor is around 200 m and the system needs to be able to distinguish two targets that are 1 meter apart. From these requirements, we can compute the waveform parameters.

```
fc = 77e9;  
c = 3e8;  
lambda = c/fc;
```

The sweep time can be computed based on the time needed for the signal to travel the unambiguous maximum range. In general, for an FMCW radar system, the sweep time should be at least five to six times the round-trip time. This example uses a factor of 5.5.

```
range_max = 200;  
tm         = 5.5 * range2time(range_max,c);
```

The sweep bandwidth can be determined according to the range resolution and the sweep slope is calculated using both sweep bandwidth and sweep time.

```
range_res    = 1;  
bw           = rangeres2bw(range_res,c);  
sweep_slope  = bw/tm;
```

Because an FMCW signal often occupies a large bandwidth, setting the sample rate blindly to twice the bandwidth often stresses the capability of A/D converter hardware. To address this issue, we can often choose a lower sample rate. Consider two things here:

1. For a complex sampled signal, the sample rate can be set to the same as the bandwidth.
2. FMCW radars estimate the target range using the beat frequency embedded in the dechirped signal. The maximum beat frequency the radar needs to detect is the sum of the beat frequency corresponding to the maximum range and the maximum Doppler frequency. Hence, the sample rate only needs to be twice the maximum beat frequency.

In this example, the beat frequency corresponding to the maximum range is as follows.

```
fr_max = range2beat(range_max,sweep_slope,c);
```

Because the maximum speed of a traveling car is about 230 km/h. Hence the maximum Doppler shift and the maximum beat frequency can be computed as

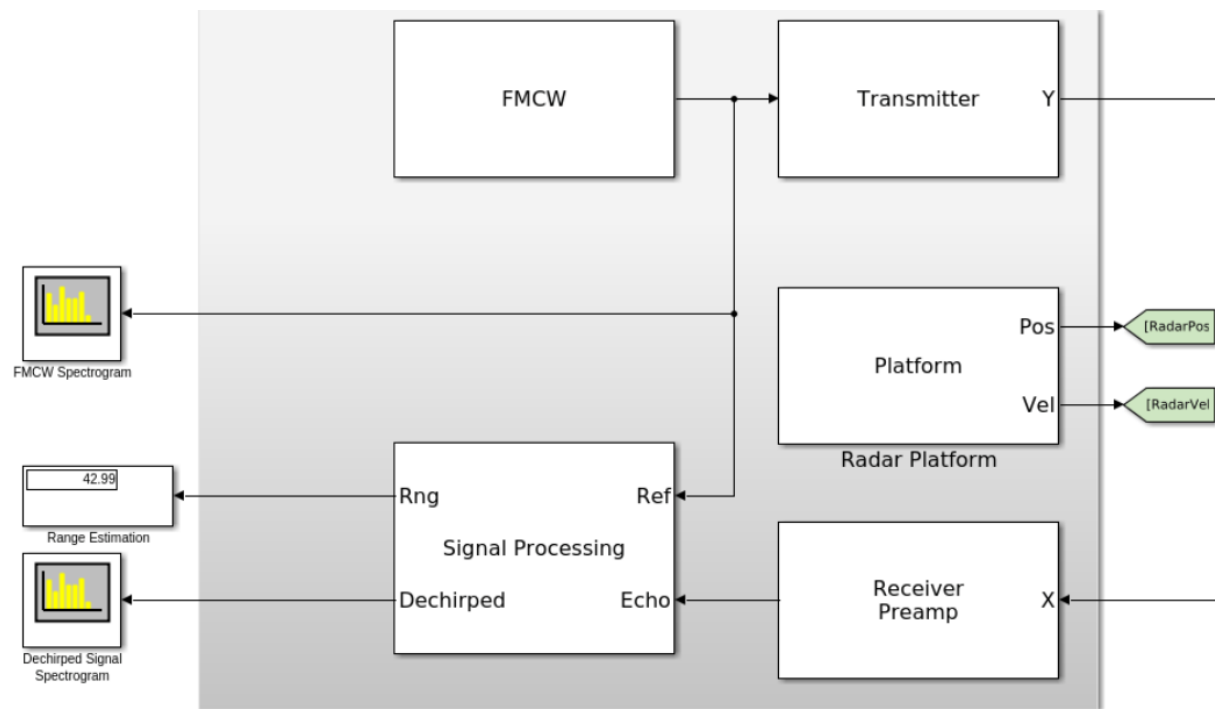
```
v_max = 230*1000/3600;
```

```
fd_max = speed2dop(2*v_max,lambda);
```

```
fb_max = fr_max+fd_max;
```

This example adopts a sample rate of the larger of twice the maximum beat frequency and the bandwidth.

```
fs = max(2*fb_max,bw);
```



RADAR

The following table summarizes the radar parameters.

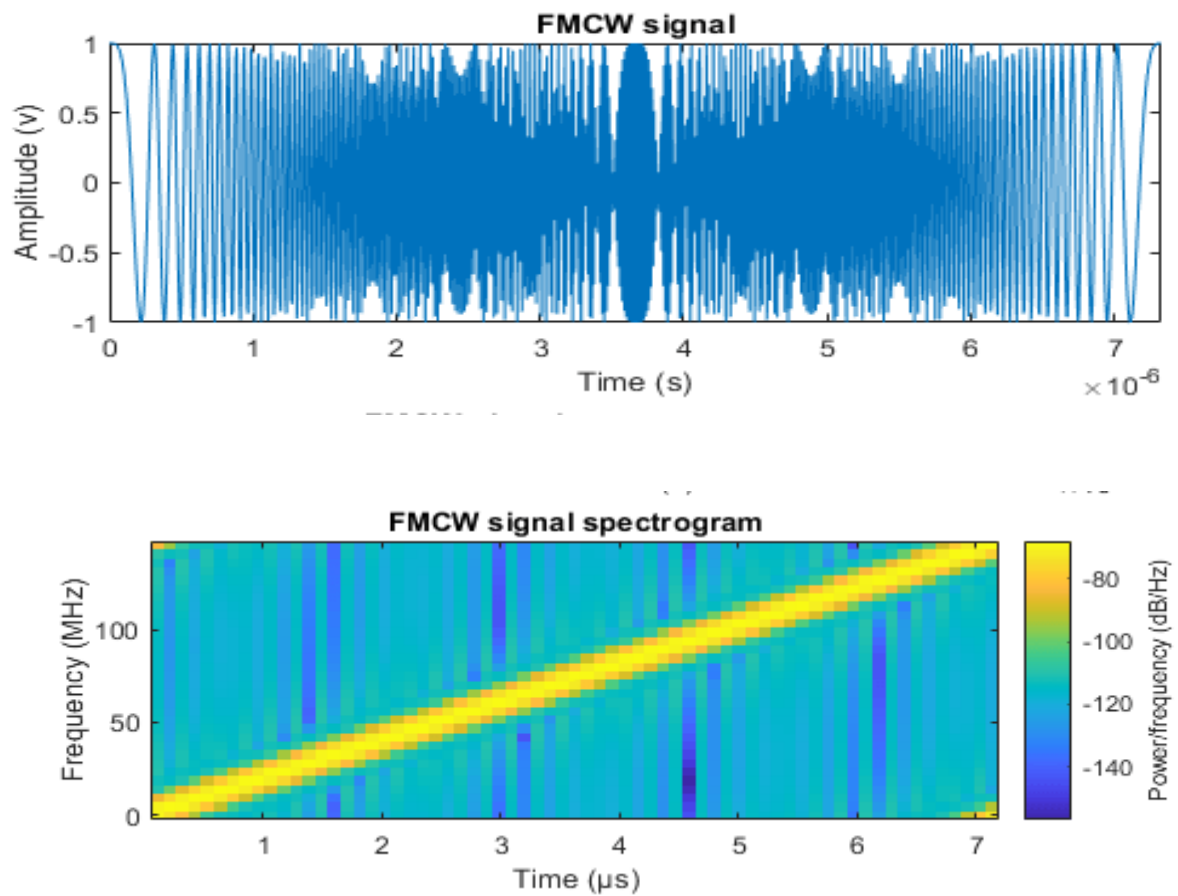
System parameters	Value
Operating frequency (GHz)	77
Maximum target range (m)	200
Range resolution (m)	1
Maximum target speed (km/h)	230
Sweep time (microseconds)	7.33
Sweep bandwidth (MHz)	150
Maximum beat frequency (MHz)	27.30
Sample rate (MHz)	150

With this information, set up the FMCW waveform used in the radar system.

```
waveform = phased.FMCWWaveform('SweepTime',tm,'SweepBandwidth',bw, ...
    'SampleRate',fs);
```

This is an up-sweep linear FMCW signal, often referred to as a sawtooth shape. Examine the time-frequency plot of the generated signal.

```
sig = waveform();
subplot(211); plot(0:1/fs:tm-1/fs,real(sig));
xlabel('Time (s)'); ylabel('Amplitude (v)');
title('FMCW signal'); axis tight;
subplot(212); spectrogram(sig,32,16,32,fs,'yaxis');
title('FMCW signal spectrogram');
```



Target Model

The target of an ACC radar is usually a car in front of it. This example assumes the target car is moving 43 m ahead of the car with the radar, at a speed of 96 km/h along the x-axis.

Compute the radar cross section of a car based on the distance between the radar and the target car.

```
car_dist = 43;
car_speed = 96*1000/3600;
car_rcs = db2pow(min(10*log10(car_dist)+5,20));
cartarget = phased.RadarTarget('MeanRCS',car_rcs,'PropagationSpeed',c,
'OperatingFrequency', fc);
carmotion = phased.Platform('InitialPosition',[car_dist;0;0.5],...
'VeLOCITY',[car_speed;0;0]);
```

Assume the propagation model to be free space.

```
channel = phased.FreeSpace('PropagationSpeed',c,...  
    'OperatingFrequency',fc,'SampleRate',fs,'TwoWayPropagation',true);
```

Radar System Setup

The rest of the radar system includes the transmitter, the receiver, and the antenna. This example uses the parameters presented in [1]. Note that this example models only main components and omits the effect from other components, such as coupler and mixer. In addition, for the sake of simplicity, the antenna is assumed to be isotropic and the gain of the antenna is included in the transmitter and the receiver.

```
ant_aperture = 6.06e-4;           % in square meter  
ant_gain = aperture2gain(ant_aperture,lambda); % in dB
```

```
tx_ppower = db2pow(5)*1e-3;      % in watts  
tx_gain = 9+ant_gain;           % in dB
```

```
rx_gain = 15+ant_gain;           % in dB  
rx_nf = 4.5;                     % in dB
```

```
transmitter = phased.Transmitter('PeakPower',tx_ppower,'Gain',tx_gain);  
receiver = phased.ReceiverPreamplifier('Gain',rx_gain,'NoiseFigure',rx_nf,...  
    'SampleRate',fs);
```

Automotive radars are generally mounted on vehicles, so they are often in motion. This example assumes the radar is traveling at a speed of 100 km/h along x-axis. So the target car is approaching the radar at a relative speed of 4 km/h.

```
radar_speed = 100*1000/3600;  
radarmotion = phased.Platform('InitialPosition',[0;0;0.5],...  
    'Velocity',[radar_speed;0;0]);
```

Radar Signal Simulation

As briefly mentioned in earlier sections, an FMCW radar measures the range by examining the beat frequency in the dechirped signal. To extract this frequency, a dechirp operation is performed by mixing the received signal with the transmitted signal. After the mixing, the dechirped signal contains only individual frequency components that correspond to the target range.

In addition, even though it is possible to extract the Doppler information from a single sweep, the Doppler shift is often extracted among several sweeps because within one pulse, the Doppler frequency is indistinguishable from the beat frequency. To measure the range and Doppler, an FMCW radar typically performs the following operations:

1. The waveform generator generates the FMCW signal.
2. The transmitter and the antenna amplify the signal and radiate the signal into space.
3. The signal propagates to the target, gets reflected by the target, and travels back to the radar.
4. The receiving antenna collects the signal.
5. The received signal is dechirped and saved in a buffer.
6. Once a certain number of sweeps fill the buffer, the Fourier transform is performed in both range and Doppler to extract the beat frequency as well as the Doppler shift. can then estimate the range and speed of the target using these results. Range and Doppler can also be shown as an image and give an intuitive indication of where the target is in the range and speed domain.

The next section simulates the process outlined above. A total of 64 sweeps are simulated and a range Doppler response is generated at the end.

During the simulation, a spectrum analyzer is used to show the spectrum of each received sweep as well as its dechirped counterpart.

```
specanalyzer = spectrumAnalyzer('SampleRate',fs, ...  
    'Method','welch','AveragingMethod','running', ...  
    'PlotAsTwoSidedSpectrum',true, 'FrequencyResolutionMethod','rbw', ...  
    'Title','Spectrum for received and dechirped signal', ...  
    'ShowLegend',true);
```

Next, run the simulation loop.

```
rng(2012);
NswEEP = 64;
xr = complex(zeros(waveform.SampleRate*waveform.SweepTime,NswEEP));

for m = 1:NswEEP
    % Update radar and target positions
    [radar_pos,radar_vel] = radarmotion(waveform.SweepTime);
    [tgt_pos,tgt_vel] = carmotion(waveform.SweepTime);

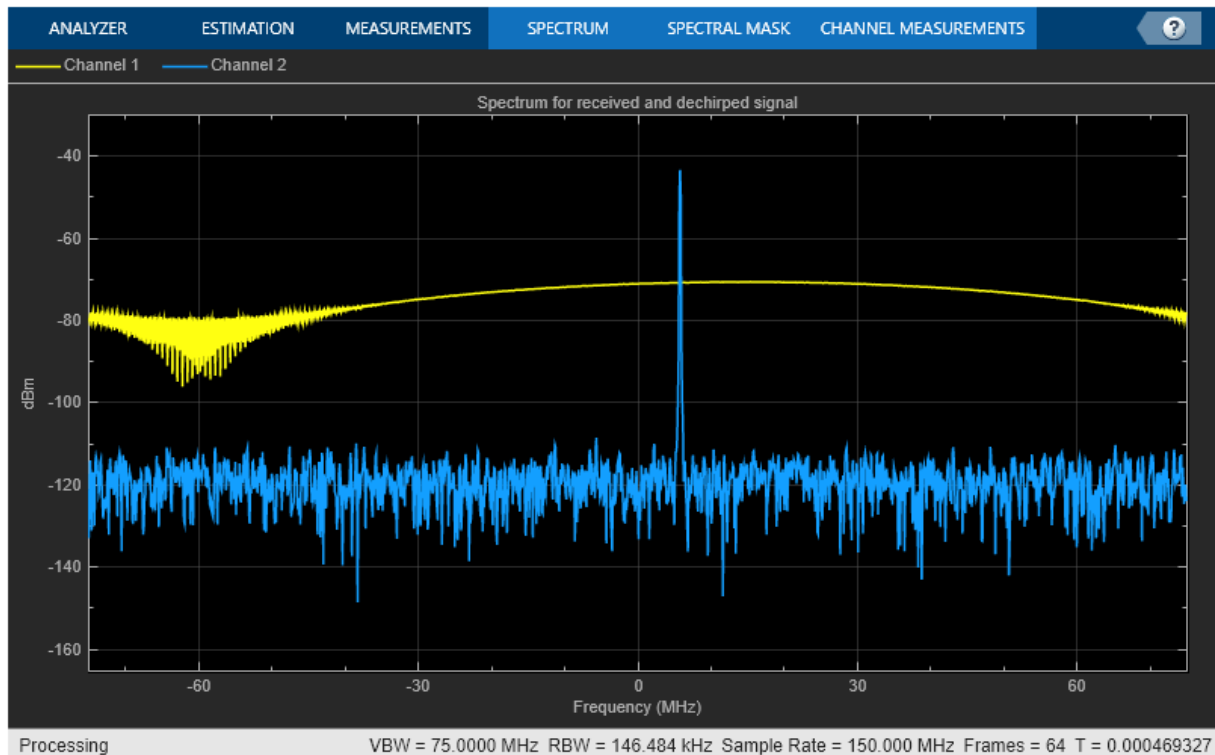
    % Transmit FMCW waveform
    sig = waveform();
    txsig = transmitter(sig);

    % Propagate the signal and reflect off the target
    txsig = channel(txsig,radar_pos,tgt_pos,radar_vel,tgt_vel);
    txsig = cartarget(txsig);

    % Dechirp the received radar return
    txsig = receiver(txsig);
    dechirpsig = dechirp(txsig,sig);

    % Visualize the spectrum
    specanalyzer([txsig dechirpsig]);

    xr(:,m) = dechirpsig;
end
```

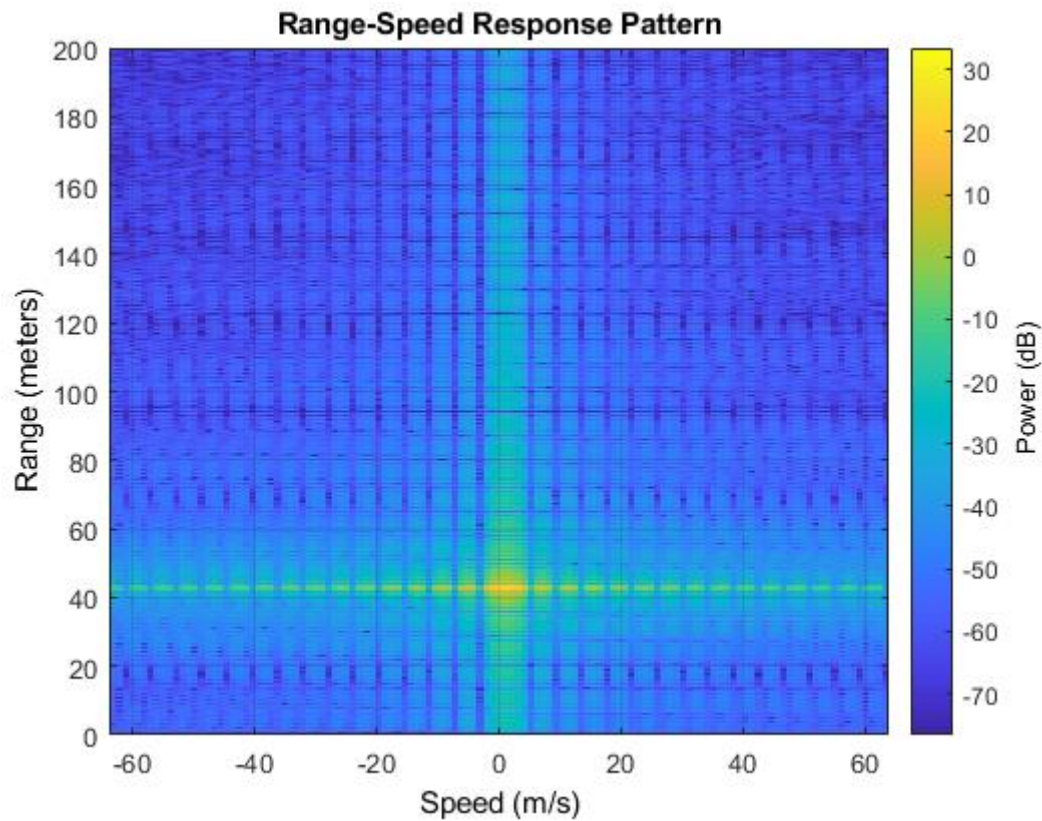


Range and Doppler Estimation

Before estimating the value of the range and Doppler, take a look at the zoomed range Doppler response of all 64 sweeps.

```
rngdopresp = phased.RangeDopplerResponse('PropagationSpeed',c,...
    'DopplerOutput','Speed','OperatingFrequency',fc,'SampleRate',fs,...
    'RangeMethod','FFT','SweepSlope',sweep_slope,...
    'RangeFFTLengthSource','Property','RangeFFTLength',2048,...
    'DopplerFFTLengthSource','Property','DopplerFFTLength',256);
```

```
clf;
plotResponse(rngdopresp,xr);           % Plot range Doppler map
axis([-v_max v_max 0 range_max])
clim = caxis;
```



From the range Doppler response, we can see that the car in front is a bit more than 40 m away and appears almost static. This is expected because the radial speed of the car relative to the radar is only 4 km/h, which translates to a mere 1.11 m/s.

There are many ways to estimate the range and speed of the target car. For example, we can choose almost any spectral analysis method to extract both the beat frequency and the Doppler shift. This example uses the root MUSIC algorithm to extract both the beat frequency and the Doppler shift.

As a side note, although the received signal is sampled at 150 MHz so the system can achieve the required range resolution, after the dechirp, we need to sample it only at a rate that corresponds to the maximum beat frequency. Since the maximum beat frequency is in general less than the required sweeping bandwidth, the signal can be decimated to alleviate the hardware cost. The following code shows the decimation process.

```
Dn = fix(fs/(2*fb_max));
for m = size(xr,2):-1:1
    xr_d(:,m) = decimate(xr(:,m),Dn,'FIR');
```


end

fs_d = fs/Dn;

To estimate the range, firstly, estimate the beat frequency using the coherently integrated sweeps and then converted to the range.

fb_rng = rootmusic(pulsint(xr_d,'coherent'),1,fs_d);

rng_est = beat2range(fb_rng,sweep_slope,c)

rng_est =

42.9976

Second, estimate the Doppler shift across the sweeps at the range where the target is present.

peak_loc = val2ind(rng_est,c/(fs_d*2));

fd = -rootmusic(xr_d(peak_loc,:),1,1/tm);

v_est = dop2speed(fd,lambda)/2

v_est =

1.0830

Note that both range and Doppler estimation are quite accurate.

Range Doppler Coupling Effect

One issue associated with linear FM signals, such as an FMCW signal, is the range Doppler coupling effect. As discussed earlier, the target range corresponds to the beat frequency. Hence, an accurate range estimation depends on an accurate estimate of beat frequency. However, the presence of Doppler shift changes the beat frequency, resulting in a biased range estimation.

For the situation outlined in this example, the range error caused by the relative speed between the target and the radar is as follows

deltaR = rdcoupling(fd,sweep_slope,c)

deltaR = -0.0041

This error is so small that we can safely ignore it.

Even though the current design achieves the desired performance, one parameter warrants further attention. In the current configuration, the sweep time is about 7 microseconds. Therefore, the system needs to sweep a 150 MHz band within a very short period. Such an automotive radar might not be able to meet the cost requirement. Besides, given the velocity of a car, there is no need to make measurements every 7 microseconds. Hence, automotive radars often use a longer sweep time. For example, the waveform used in [2] has the same parameters as the waveform designed in this example except a sweep time of 2 ms.

A longer sweep time makes the range Doppler coupling more prominent. To see this effect, first reconfigure the waveform to use 2 ms as the sweep time.

```

waveform_tr = clone(waveform);
release(waveform_tr);
tm = 2e-3;
waveform_tr.SweepTime = tm;
sweep_slope = bw/tm;
Now calculate the range Doppler coupling.

deltaR = rdcoupling(fd,sweep_slope,c)
deltaR = -1.1118

```

A range error of 1.14 m can no longer be ignored and needs to be compensated. Naturally, we might consider doing so following the same procedure outlined in earlier sections, estimating both range and Doppler, figuring out the range Doppler coupling from the Doppler shift, and then remove the error from the estimate.

Unfortunately this process does not work very well with the long sweep time. The longer sweep time results in a lower sampling rate across the sweeps, thus reducing the capability of the radar for unambiguously detecting high speed vehicles. For instance, using a sweep time of 2 ms, the maximum unambiguous speed the radar system can detect using the traditional Doppler processing is as follows

```

v_unambiguous = dop2speed(1/(2*tm),lambda)/2
v_unambiguous = 0.4870

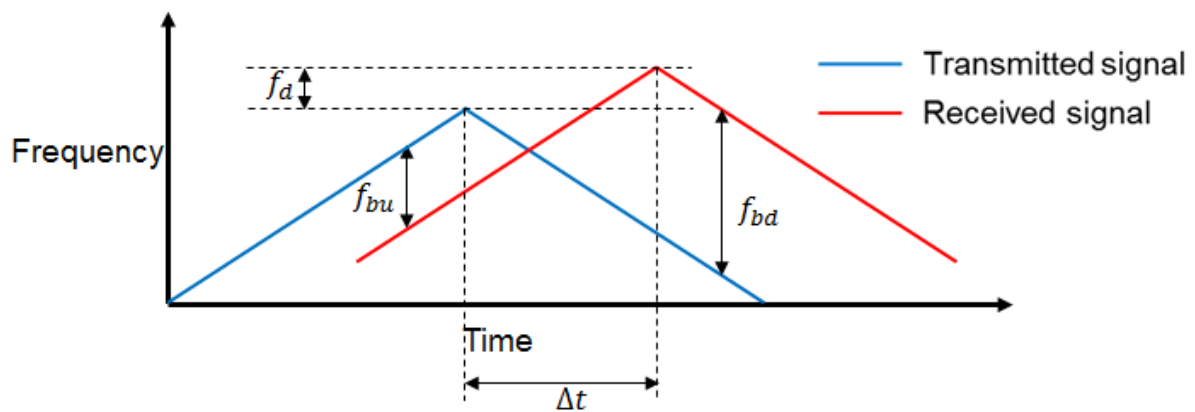
```

The unambiguous speed is only 0.48 m/s, which means that the relative speed, 1.11 m/s, cannot be unambiguously detected. This means that not only the target car will appear slower in Doppler processing, but the range Doppler coupling also cannot be correctly compensated.

One way to resolve such ambiguity without Doppler processing is to adopt a triangle sweep pattern. Next section shows how the triangle sweep addresses the issue.

Triangular Sweep

In a triangular sweep, there is one up sweep and one down sweep to form one period, as shown in the following figure.



The two sweeps have the same slope except different signs. From the figure, we can see that the presence of Doppler frequency, f_d , affects the beat frequencies (f_{bu} and f_{bd}) differently in up and down sweeps. Hence by combining the beat frequencies from both up and down sweep, the coupling effect from the Doppler can be averaged out and the range estimate can be obtained without ambiguity.

First, set the waveform to use triangular sweep.

```
waveform_tr.SweepDirection = 'Triangle';
```

Now simulate the signal return. Because of the longer sweep time, fewer sweeps (16 vs. 64) are collected before processing.

```
Nsweep = 16;
```

```
xr = helperFMCWSimulate(Nsweep,waveform_tr,radarmotion,carmotion,...  
    transmitter,channel,cartarget,receiver);
```

The up sweep and down sweep are processed separately to obtain the beat frequencies corresponding to both up and down sweep.

```
fbu_rng = rootmusic(pulsint(xr(:,1:2:end),'coherent'),1,fs);
```

```
fbd_rng = rootmusic(pulsint(xr(:,2:2:end),'coherent'),1,fs);
```

Using both up sweep and down sweep beat frequencies simultaneously, the correct range estimate is obtained.

```
rng_est = beat2range([fbu_rng fbd_rng],sweep_slope,c)
```

```
rng_est =
```

```
42.9658
```

Moreover, the Doppler shift and the velocity can also be recovered in a similar fashion.

```
fd = -(fbu_rng+fbd_rng)/2;
```

```
v_est = dop2speed(fd,lambda)/2
```

```
v_est = 1.1114
```

The estimated range and velocity match the true values, 43 m and 1.11 m/s, very well.

Future scope and applications

The successful implementation of FMCW radar in the adaptive cruise control application opens up several future scope and potential applications. Some of them are:

1. Collision Avoidance Systems: FMCW radar can be further developed to enable collision avoidance systems in vehicles. By continuously monitoring the surrounding environment and detecting potential collisions, the radar system can provide timely warnings or even automatically initiate braking or evasive maneuvers.

2. Traffic Management: FMCW radar can be utilized for intelligent traffic management systems. By integrating radar sensors into traffic infrastructure, such as traffic lights or road signs, the radar system can collect real-time data on traffic flow, congestion, and vehicle movement patterns. This information can be used to optimize traffic management strategies and improve overall traffic efficiency.

3. Object Detection and Tracking: FMCW radar's ability to accurately estimate range and velocity makes it suitable for object detection and tracking applications beyond automotive

use. It can be employed in areas such as surveillance systems, unmanned aerial vehicles (UAVs), or industrial automation to detect and track objects or monitor their movement.

4. Robotics and Autonomous Systems: FMCW radar can play a crucial role in robotics and autonomous systems by providing reliable perception capabilities. It can assist in mapping and environment perception tasks for autonomous vehicles, drones, or robots operating in various settings, including industrial environments, warehouses, or outdoor terrains.

5. Weather Monitoring: FMCW radar's ability to measure range and velocity can be leveraged for weather monitoring applications. By analyzing the reflections and Doppler shifts from precipitation particles, FMCW radar can provide valuable information about weather conditions, such as rainfall intensity, wind speed, and storm tracking, aiding in weather forecasting and early warning systems.

Conclusion

In conclusion, this project demonstrates the effective utilization of FMCW radar signals for range and Doppler estimation in an automotive automatic cruise control system. By generating a range Doppler map from the received signals, the radar system can accurately determine the distance and relative velocity between the host vehicle and other vehicles in its vicinity. The use of a triangle sweep enables compensation for the range Doppler coupling effect, ensuring accurate estimation results. Additionally, the project highlights the impact of multipath propagation on the signal level and discusses its potential implications. Overall, the implementation of FMCW radar in automatic cruise control showcases its potential for enhancing safety and improving driving experiences in the automotive industry.

References

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